

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12405678
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	Yamamura; Yoshihiro et al.

Input device and biometric information detection device

Abstract

An input device configured to communicate with an information processing device includes a light source unit configured to emit light; a case including a first surface including a first opening and configured to emit the light toward a target object through the first opening, and a second surface including a second opening and configured to emit the light toward a living body through the second opening; a first light reception unit configured to receive first light from the target object through the first opening of the first surface; and a second light reception unit configured to receive second light from the living body through the second opening of the second surface.

Inventors: Yamamura; Yoshihiro (Azumino, JP), Yamamoto; Hidetoshi (Suwa, JP), Sasahara; Hideo (Suwa, JP), Ozawa; Yuya (Azumino, JP)

Applicant: SEIKO EPSON CORPORATION (Tokyo, JP)

Family ID: 92700225

Assignee: SEIKO EPSON CORPORATION (Tokyo, JP)

Appl. No.: 18/603235

Filed: March 13, 2024

Prior Publication Data

Document Identifier	Publication Date
US 20240310930 A1	Sep. 19, 2024

Foreign Application Priority Data

JP	2023-039390	Mar. 14, 2023
----	-------------	---------------

Publication Classification

Int. Cl.: G06F3/0354 (20130101); **A61B5/00** (20060101); **A61B5/024** (20060101)

U.S. Cl.:

CPC G06F3/03543 (20130101); **A61B5/02427** (20130101); **A61B5/6897** (20130101);

Field of Classification Search

CPC: G06F (3/033); G06F (3/0386); G06F (3/0383); G06F (2203/04807); G06F (3/03543); G06F (3/03545); G06F (3/03541); G06F (3/04883); G06F (3/03542); G06F (3/03549); G06F (3/0312); G06F (21/32); G06F (3/042); G06V (40/1341)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2007/0211033	12/2006	Farag	345/163	G06F 3/03543
2008/0284735	12/2007	Shim	345/166	G06F 3/0317
2013/0113705	12/2012	Gu	345/166	G06F 3/03543

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2008204388	12/2007	JP	N/A

Primary Examiner: Patel; Sanjiv D.

Attorney, Agent or Firm: JCIPRNET

Background/Summary

(1) The present application is based on, and claims priority from JP Application Serial Number 2023-039390, filed Mar. 14, 2023, the disclosure of which is hereby incorporated by reference herein in its entirety.

BACKGROUND

1. Technical Field

(2) The present disclosure relates to an input device and a biometric information detection device.

2. Related Art

(3) As an optical mouse, there hitherto has been proposed an optical mouse including a mode selection switch for performing selection between a mouse mode and a pulse measurement mode. In the mouse mode, a moving direction of the mouse and a position thereof are detected. In the pulse measurement mode, various types of biometric information (a pulse rate, a vascular age, and the like) are calculated.

(4) However, an optical mouse disclosed in JP-A-2008-204388, it is difficult to acquire light from a mouse pad and light from a living body at the same time.

(5) The optical mouse includes one opening portion through which light from a mouse pad or a finger is incident so that a light reception unit receives the light. As a result, the light from the living body cannot be received while receiving the light from the mouse pad, and the light from the mouse pad cannot be received while receive the light from the living body, which causes a

problem. Thus, a mouse capable of detecting a moving direction of the mouse and biometric information at the same time has been required.

SUMMARY

(6) According to one aspect of the present disclosure, provided is an input device configured to communicate with an information processing device, the input device including: a light source unit configured to emit light, a case including a first surface including a first opening and configured to emit the light toward a target object through the first opening, and a second surface including a second opening and configured to emit the light toward a living body through the second opening, a first light reception unit configured to receive first light from the target object through the first opening of the first surface, and a second light reception unit configured to receive second light from the living body through the second opening of the second surface.

(7) According to one aspect of the present disclosure, provided is a biometric information detection device including the input device described above, and the information processing device configured to communicate with the input device, in which the input device is movably provided, and a control unit is provided, the control unit being configured to detect a moving direction of the input device, based on reception of the first light at the first light reception unit, and detect biometric information about a user, based on reception of the second light at the second light reception unit.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a plan view illustrating a schematic configuration of an optical mouse according to a first embodiment.

(2) FIG. 2 is a cross-sectional view taken along a front-and-rear direction of a mouse main body of the optical mouse in FIG. 1.

(3) FIG. 3 is a diagram illustrating a lower surface of the mouse main body.

(4) FIG. 4 is a side view of the mouse main body as viewed from the left side.

(5) FIG. 5 is a plan view of a mouse main body in a first modification example of the first embodiment.

(6) FIG. 6 is a plan view illustrating a schematic configuration of a mouse main body of an optical mouse according to a second embodiment.

(7) FIG. 7 is a plan view illustrating a schematic configuration of a mouse main body of an optical mouse according to a third embodiment.

(8) FIG. 8 is a diagram illustrating a configuration of a second light reception unit of the mouse main body illustrated in FIG. 7.

(9) FIG. 9 is a diagram illustrating a configuration of a second light reception unit of a mouse main body in a second modification example of the third embodiment.

DESCRIPTION OF EMBODIMENTS

(10) With reference to the drawings, an embodiment of the present disclosure is described.

(11) Note that, in the following drawings, the dimensions of some components may be scaled differently for ease of understanding for the components.

First Embodiment

(12) As illustrated in FIG. 1 and FIG. 2, an optical mouse **1** (biometric information detection device) according to the first embodiment includes a mouse main body **10** (input device) capable of performing a function of detecting a moving direction and a position of the mouse itself, based on reflection light from a target object P (a normal mouse function) and a function of detecting biometric information D, based on reflection light from a living body M, at the same time during the use of the optical mouse **1**.

- (13) The target object P is an upper surface of a mouse pad (omitted in illustration) on a desk, a top plate of a desk, or the like. The living body M is a fingertip (the first embodiment) or a palm of a hand (a modification example described later) of a user, or the like. Further, the biometric information D is biometric information about a user who uses the optical mouse **1**, such as a pulse rate, a vascular age, or the like of a user.
- (14) The optical mouse **1** includes the mouse main body **10**, an information processing device **11** that communicates with the mouse main body **10**, and a control unit **12** that controls the mouse main body **10**.
- (15) The mouse main body **10** is provided to be movable by an operation of a user. The mouse main body **10** includes a light source unit **20** that emits light L1, a case **30** that forms an outer shell of the mouse main body **10**, and a pair of light reception units **41** and **42** that receive light obtained by reflecting the light L1 from the light source unit **20** by the target object P and the living body M, respectively.
- (16) Here, the case **30** of the mouse main body **10** is formed to have a substantially elliptical shape vertically elongated in a front-and-rear direction in plan view. The vertical direction is defined as a longitudinal direction Y, and the horizontal direction is defined as a short direction X. Note that the direction in the short direction X may also be referred to as a right-and-left direction. In plan view in the right-and-left direction, the right side is denoted with a reference sign X1, and the left side is denoted with a reference sign X2.
- (17) The case **30** includes a lower case unit **31** including a lower surface **31a** (first surface) and an upper case unit **32** including a right surface **32b** on one end side in the short direction X (the right side X1 in the present embodiment), a left surface **32a** (second surface) on the other end side in the short direction X (the left side X2 in the present embodiment), and an upper surface **32c**. The upper case unit **32** is provided to cover the lower surface **31a** of the lower case unit **31**. The upper case unit **32** includes a movable portion **321** being movable at substantially the center portion in the short direction X on one end side (front side Y1) in the longitudinal direction Y.
- (18) The lower surface **31a** includes a first opening **30A**, and is a surface from which the light L1 is emitted toward the target object P through the first opening **30A**. The lower surface **31a** is a surface of the case **30** that comes into contact with the target object P and does not come into contact with the living body M. The lower surface **31a** is a flat surface including the longitudinal direction Y and the short direction X. As illustrated in FIG. 3, the range of the lower surface **31a** matches with a first region **31A** facing the target object P.
- (19) The first opening **30A** is only required to be arranged in the first region **31A**. In the present embodiment, the first opening **30A** is arranged at the center of the lower surface **31a** in the short direction X, slightly towards the rear side Y2 (user side, front side) with respect to the center in the longitudinal direction Y. The first opening **30A** has a rectangular shape as viewed from the lower surface side. Note that the size and the shape of the first opening **30A** may be set freely.
- (20) The left surface **32a** includes a second opening **30B**, and is a surface from which the light L1 is emitted toward the living body M through the second opening **30B**. The left surface **32a** is a surface of the case **30** that comes into contact with the living body M and does not come into contact with the target object P. The left surface **32a** forms a protruding curved surface that protrudes to the left side X2 at the center in the longitudinal direction in plan view. Note that the shape of the left surface **32a** may be set freely in accordance with usability for a user. The shape of the right surface **32b** is also set freely. As illustrated in FIG. 4, the left surface **32a** includes a second region **32A** with which a thumb (the living body M) overlaps when a user holds the case **30** with the right hand.
- (21) The second opening **30B** is only required to be arranged in the second region **32A**. In the present embodiment, the second opening **30B** is arranged at the center of the left surface **32a** in the longitudinal direction Y and the center in the up-and-down direction. In other words, the second opening **30B** is provided at a position covered with a hand or a finger of a user who uses the mouse

and holds the mouse main body **10**. The second opening **30B** has a rectangular shape as viewed from the lower surface side. Note that the size and the shape of the second opening **30B** may be set freely.

(22) As illustrated in FIG. **1**, the mouse main body **10** includes, in the case **30**, the light source unit **20** described above, the pair of openings described above (the first opening **30A** and the second opening **30B**), the first light reception unit **41** corresponding to the first opening **30A**, the second light reception unit **42** corresponding to the second opening **30B**, a deflection mirror **51** (deflection unit) arranged between the light source unit **20** and each of the pair of openings **30A** and **30B**, and a condensing lens **52** (optical member) provided between the light source unit **20** and the deflection mirror **51**.

(23) The light source unit **20** includes a light-emitting diode. The light **L1** from the light source unit **20** is emitted toward the deflection mirror **51**. The condensing lens **52** condenses the light **L1** from the light source unit **20** so that the light **L1** arrives at the deflection mirror **51**.

(24) As the deflection mirror **51**, a half mirror is adopted. The deflection mirror **51** deflects first light **L11**, which is one part of the emission light **L1** emitted from the light source unit **20**, toward the first opening **30A** provided in the lower surface **31a** of the lower case unit **31**, and transmits second light **L12**, which is another part of the emission light **L1**, and emits the second light **L12** toward the second opening **30B** provided in the left surface **32a** of the upper case unit **32**. The first light **L11** reflected and deflected by the deflection mirror **51** is reflected by the target object **P** through the first opening **30A**, and then first reflection light **L21** thus generated is incident on the first light reception unit **41**. The second light **L12** transmitted through the deflection mirror **51** is reflected by the living body **M** (a finger of a user who uses the mouse main body **10**) through the second opening **30B**, and then second reflection light **L22** thus generated is incident on the second light reception unit **42**.

(25) Orientation (deflection angle) of the deflection mirror **51** is set in accordance with conditions such as a position of the light source unit **20**, an incident angle of the emission light **L1**, orientation of the first light **L11** to be deflected, and orientation of the second light **L12** to be transmitted.

(26) The first light reception unit **41** receives the first reflection light **L21** from the target object **P** through the first opening **30A** of the lower surface **31a** of the lower case unit **31**. The second light reception unit **42** receives the second reflection light **L22** from the living body **M** through the second opening **30B** of the left surface **32a** of the upper case unit **32**.

(27) The control unit **12** is provided in the case **30**. The control unit **12** performs control for detecting the moving direction and the present position of the mouse main body **10** itself, based on a speckle pattern of the first reflection light **L21** incident on the first light reception unit **41**, and control for detecting the biometric information **D** such as a blood oxygenation level or the like of a user by using intensity of the second reflection light **L22** incident on the second light reception unit **42**. When the second light reception unit **42** receives the second reflection light **L22**, the control unit **12** performs control so that the moving direction and the current position of the mouse main body **10** is detected based on reception of the first reflection light **L21** by the first light reception unit **41**. At the same time, the control unit **12** performs control so that the biometric information **D** is detected based on reception of the second reflection light **L22** by the second light reception unit **42**.

(28) Further, when the first light reception unit **41** receives the first reflection light **L21**, and the second light reception unit **42** does not receive the second reflection light **L22**, the control unit **12** performs control for turning off the power of the optical mouse **1** so as to prevent the light source unit **20** from emitting the emission light **L1**. When the optical mouse **1** is not in use, and the power is in an ON state, such control can prevent leakage of the emission light **L1** (the second light **L12**), which is emitted from the light source unit **20** and is not reflected by a finger closing the second opening **30B**, to the outside.

(29) Further, the control unit **12** transmits information relating to the biometric information **D** about

the user, which is detected by the second reflection light L22 received by the second light reception unit 42, to the information processing device 11. The information processing device 11 is provided outside of the mouse main body 10. In the information processing device 11, processing is executed as appropriate, such as accumulation or display of the biometric information D received from the control unit 12.

(30) Further, the mouse main body 10 includes an acceleration sensor 53 (detection member) that detects movement of the mouse main body 10. The acceleration sensor 53 is embedded in the case 30.

(31) In the optical mouse 1 thus configured, when the power of the mouse main body 10 is turned on, the emission light L1 is emitted from the light source unit 20. When the second light reception unit 42 does not receive the second reflection light L22, the control unit 12 turned off the power to stop emission of the emission light L1 from the light source unit 20. When the second light reception unit 42 receives the second reflection light L22, control is performed so that the moving direction and the current position of the mouse main body 10 is detected based on reception of the first reflection light L21 by the first light reception unit 41. At the same time, control is performed so that the biometric information D is detected based on reception of the second reflection light L22 by the second light reception unit 42.

(32) Further, when the acceleration sensor 53 detects movement of the mouse main body 10, the control unit 12 performs control for detecting the biometric information D about the user, based on the second reflection light L22 received by the second light reception unit 42. After that, in a step in which the biometric information D is detected by receiving the light by the second light reception unit 42, control is performed so that a heart rate sensor (PPG sensor) or the like provided to the control unit 12, which is omitted in illustration, is continuously turned ON and the PPG sensor is turned OFF when the biometric information D is not detected.

(33) The biometric information D detected by the control unit 12 is transmitted to the information processing device 11 provided to the outside of the mouse main body 10. Then, the information processing device 11 subjects the received biometric information D to data processing as appropriate. In other words, the biometric information D received from the control unit 12 is monitored through, for example, display by a monitor of a personal computer or a mobile terminal, which is omitted in illustration. With this, a physical condition of a user can be recognized. For example, fluctuations or cessation in a pulse wave of a user, vessel constriction due to excessing stress, or the like can be monitored in real time, which enables proactive measures for prevention.

(34) Next, actions of the mouse main body 10 and the optical mouse 1 are described.

(35) As illustrated in FIG. 1, the mouse main body 10 according to the present embodiment communicates with the information processing device 11. The mouse main body 10 includes the light source unit 20 that emits the emission light L1, the case 30 that includes the lower surface 31a including the first opening 30A and that emits the light toward the target object P through the first opening 30A, and the left surface 32a including the second opening 30B and that emits the light toward the living body M through the second opening 30B, the first light reception unit 41 that receives the first reflection light L21 from the target object P through the first opening 30A of the lower surface 31a, and the second light reception unit 42 that receives the second reflection light L22 from the living body M through the second opening 30B of the left surface 32a.

(36) Therefore, in the present embodiment, the emission light L1 emitted from the light source unit 20 can be reflected by the target object P and the living body M through the first opening 30A and the second opening 30B, respectively. In the present embodiment, the first opening 30A through which the target object P is irradiated with the first light L11 and the second opening 30B through which the living body M is irradiated with the second light L12 are provided. Thus, there is no need to perform switching between detection of the target object P and detection of the living body M. The reflection light L21 and the reflection light L22 obtained by irradiating both the objects (the target object P and the living body M) with the light L1 and capturing its reflection therefrom are

received by the first light reception unit **41** and the second light reception unit **42**, respectively at the same time. The moving direction and the position of the mouse main body **10** can be detected from the target object P, and the biometric information D about the user can be detected from the living body M, at the same time. In other words, even when the second light reception unit **42** receives the second reflection light L22, the first light reception unit **41** is capable of receiving the first reflection light L21.

(37) Further, in the present embodiment, the mouse main body **10** is communicable with the information processing device **11**. Thus, the biometric information D acquired by the mouse main body **10** can be transmitted to the information processing device **11**. Then, processing can be executed as appropriate, such as display of the acquired biometric information D on a terminal, accumulation of the data, and comparison.

(38) Further, in the present embodiment, the deflection mirror **51** is included. The deflection mirror **51** deflects one part of the emission light L1 (the first light L11), and transmits another part of the emission light L1 (the second light L12). The deflection mirror **51** deflects the first light L11 toward the first opening **30A** of the lower surface **31a**, and transmits the second light L12 toward the second opening **30B** of the left surface **32a**. In this case, the deflection mirror **51** is used. With this, one emission light L1 can be formed into the two parts of light (the first light L11 and the second light L12) advancing to the target object P and the living body M, respectively. With this simple configuration, the moving direction of the mouse main body **10** and the biometric information D can be detected at the same time.

(39) In the mouse main body **10** according to in the present embodiment, the lower surface **31a** is a surface of the case **30** that comes into contact with the target object P and does not come into contact with the living body M. The left surface **32a** is a surface of the case **30** that comes into contact with the living body M and does not come into contact with the target object P. In this case, the lower surface **31a** that comes into contact with the target object P and includes the first opening **30A** and the left surface **32a** that comes into contact with the living body M and includes the second opening **30B** are different surfaces, and do not come into contact with the targets (the target object P and the living body M) that are not detected by the respective surfaces. With this, detection can be performed at high accuracy.

(40) In the mouse main body **10** according to the present embodiment, the case **30** includes the lower case unit **31** including the lower surface **31a** and the upper case unit **32** covering the lower surface **31a** and including the left surface **32a**. The upper case unit **32** includes the left surface **32a** at the one end side of the case **30** in the short direction X. In this case, the second opening **30B** can be formed in the left surface **32a** of the upper case unit **32**. Thus, when a user holds and uses the mouse main body **10**, a finger of the user is placed on the second opening **30B**. With this, the biometric information D can be detected from the finger.

(41) The mouse main body **10** according to the present embodiment includes the condensing lens **52** between the light source unit **20** and the deflection mirror **51**. In this case, the emission light L1 emitted from the light source unit **20** can be condensed by the condensing lens **52**, and can be incident on the deflection mirror **51**. In other words, the emission light L1 can be efficiently incident on the deflection mirror **51** without a loss, which is effective in the present embodiment where the deflection mirror **51** deflects one part of the light, and transmits another part of the light.

(42) The optical mouse **1** according to the present embodiment includes the mouse main body **10** described above and the information processing device **11** that communicates with the mouse main body **10**. The mouse main body **10** is provided to be movable. The control unit **12** is included. The control unit **12** detects the moving direction of the mouse main body **10**, based on reception of the first reflection light L21 by the first light reception unit **41**, and detects the biometric information D about the user, based on reception of the second reflection light L22 by the second light reception unit **42**. In this case, the control unit **12** is capable of controlling detection timing for the moving direction and the position of the mouse main body **10** and the biometric information D, the

ON/OFF states, and the like in accordance with the light reception states of the first light reception unit **41** and the second light reception unit **42**.

(43) With the optical mouse **1** according to the present embodiment, when the second light reception unit **42** receives the second reflection light **L22**, the control unit **12** detects the moving direction of the mouse main body **10**, based on reception of the first reflection light **L21** by the first light reception unit **41**. In this case, when the second light reception unit **42** does not receive the second reflection light **L22**, the living body **M** is not located on the second opening **30B**, in other words, a user does not operate the mouse main body **10**. Thus, the control unit **12** is capable of performing control for stopping detection the moving direction and the position of the mouse main body **10** by the first light reception unit **41**.

(44) Further, in the present embodiment, the control unit **12** transmits the information relating to the biometric information **D** about the user to the information processing device **11**. In this case, the information processing device **11** can execute processing as appropriate, such as display the biometric information **D**, which is received and acquired from the mouse main body **10**, on a terminal, accumulation of the data, and comparison.

(45) Further, in the present embodiment, the acceleration sensor **53** that detects movement of the mouse main body **10** is included. When the acceleration sensor **53** detects movement of the mouse main body **10**, the control unit **12** detects the biometric information **D** about the user, based on reception of the second reflection light **L22** by the second light reception unit **42**. In this case, the biometric information **D** can be acquired only when the acceleration sensor **53** recognizes movement of the mouse main body **10**. Thus, when the mouse main body **10** does not move, the mouse main body **10** is not operated, and hence emission of the second light **L12** from the second opening **30B** to the outside can be suppressed.

(46) According to at least one embodiment described above, the moving direction of the mouse main body **10** and the biometric information **D** can be detected at the same time.

First Modification Example

(47) FIG. **5** is a plan view of a mouse main body **10A** of an optical mouse **1A** (biometric information detection device) in a first modification example of the first embodiment.

(48) As illustrated in FIG. **5**, in the first modification example, the case **30** of the mouse main body **10A** includes the lower case unit **31** including the lower surface **31a** and the upper case unit **32** covering the lower surface **31a** and including the upper surface **32c** (second surface). The upper surface **32c** of the upper case unit **32** is provided with a second opening **30C** at the center in the short direction **X** on the rear side **Y2** in the longitudinal direction **Y**. The second opening **30C** is located in a region of the upper surface **32c** that is covered with a palm of a hand of a user. In the present modification example, similarly to the first embodiment described above, the second light transmitted through the deflection mirror is also reflected by a living body (a palm of a hand of a user) through the second opening **30C**, and then the second reflection light thus generated is also incident on the second light reception unit.

(49) In the present modification example, a user who uses the mouse main body **10A** can operate the mouse main body **10A** whether the user is right-handed or left-handed, and biometric information about the user can be detected. In other words, similarly to the first embodiment described above, in a case in which the second opening **30B** is provided in the left surface **32a** (see FIG. **1**), when the mouse main body **10** is operated by a left hand, a finger (living body) may not be placed at the position of the second opening **30B**. However, in the present modification example, the mouse main body **10A** may be held by any one of a right hand and a left hand, and hence has advantageous usability.

Second Embodiment

(50) FIG. **6** is a plan view illustrating an internal structure of a mouse main body **10B** of an optical mouse **1B** (biometric information detection device) according to a second embodiment.

(51) As illustrated in FIG. **6**, in the mouse main body **10B** according to the second embodiment,

emission light coming from a light source unit **20A** includes light having a first wavelength (red light **L1a**) and light having a second wavelength being a wavelength different from the first wavelength (blue light **L1b**). The second wavelength of the blue light **L1b** is shorter than the first wavelength of the red light **L1a**. For example, the wavelength band of the first wavelength of the red light **L1a** is near 620 nm with a red emission color, and the wavelength band of the second wavelength of the blue light **L1b** is near 460 nm with a blue emission color. In other words, the light source unit **20A** according to the second embodiment emits the light with the two emission colors of red and blue.

(52) A deflection mirror **51A** (deflection unit) according to the second embodiment deflects the blue light **L1b** toward the target object, and deflects the red light **L1a** toward the living body. The red light **L1a** is deflected toward the second opening **30B** by the deflection mirror **51A**. The red light **L1a** is reflected by the living body, and the reflection light thus generated (red reflection light **L2a**) is incident on the second light reception unit **42**. The blue light **L1b** is deflected toward the first opening **30A** by the deflection mirror **51A**. The blue light **L1b** is reflected by the target object, and the reflection light thus generated (blue reflection light **L2b**) is incident on the first light reception unit **41**.

(53) In this manner, the deflection mirror **51A** according to the second embodiment is configured by a diffraction element that diffracts light, in place of the half mirror in the first embodiment. The deflection mirror **51A** is arranged obliquely at an appropriate angle with respect to the red light **L1a** and the blue light **L1b** that are emitted from the light source unit **20A**.

(54) Note that, in the second embodiment, there is adopted a configuration in which one light source unit **20A** is provided. However, a configuration in which a plurality of light source units that respectively emit the two parts of light **L1a** and **L1b** are provided may be adopted. In other words, a light source unit that emits the blue light **L1b** for detecting the moving direction of the mouse main body **10B** and a light source unit that emits the red light **L1a** for detecting biometric information about a user may be provided.

(55) According to the second embodiment, the two types of light (the red light **L1a** and the blue light **L1b**) can be used, and the respective light can be deflected to the first opening **30A** and the second opening **30B** by the deflection mirror **51A**. The different types of light are separately adopted, and hence the light can be securely reflected by the target object **P** and the living body **M** and received by the light reception units **41** and **42**.

Third Embodiment

(56) FIG. 7 is a plan view illustrating an internal structure of a mouse main body **10C** of an optical mouse **1C** according to a third embodiment.

(57) In the mouse main body **10C** according to the third embodiment, the two parts of light having different wavelengths are incident on the second opening **30B** and the second light reception unit **42** in the configuration of the second embodiment described above. As illustrated in FIG. 7, in the mouse main body **10C** according to the third embodiment, emission light from a light source unit **20B** includes the light having the first wavelength (the red light **L1a**), the light having the second wavelength (the blue light **L1b**), and light having a third wavelength (green light **L1c**), which have wavelengths different from one another.

(58) The second wavelength of the blue light **L1b** is shorter than the first wavelength of the red light **L1a** and the third wavelength of the green light **L1c**. The third wavelength of the green light **L1c** is longer than the first wavelength of the red light **L1a**. For example, the wavelength band of the first wavelength of the red light **L1a** is near 620 nm with a red emission color, the wavelength band of the second wavelength of the blue light **L1b** is near 460 nm with a blue emission color, and the wavelength band of the third wavelength of the green light **L1c** is near 530 nm with a green emission color. For example, the wavelength band of the second wavelength of the blue light **L1b** is near 530 nm with a green emission color, the wavelength band of the first wavelength of the red light **L1a** is near 620 nm with a red emission color, and the wavelength band of the third

wavelength of the green light **L1c** is near 530 nm with a green emission color. In other words, the light source unit **20B** according to the third embodiment emit the light with the three colors of red, blue, and green.

(59) The deflection mirror **51A** (deflection unit) according to the third embodiment deflects the blue light **L1b** toward the target object **P**, and deflects the red light **L1a** and the green light **L1c** toward the living body **M**. Each of the red light **L1a** and the green light **L1c** is deflected toward the second opening **30B** by the deflection mirror **51A**. The red light **L1a** and the green light **L1c** are reflected respectively by the living body **M**, and the reflection light thus generated (the red reflection light **L2a** and the green reflection light **L2c**) are incident on the second light reception unit **42**. The blue light **L1b** is deflected toward the first opening **30A** by the deflection mirror **51A**. The blue light **L1b** is reflected by the target object **P**, and the reflection light thus generated (blue reflection light **L2b**) is incident on the first light reception unit **41**.

(60) In this manner, the deflection mirror **51A** according to the third embodiment is configured by a diffraction element that diffracts light, in place of the half mirror in the first embodiment. The deflection mirror **51A** is arranged obliquely at an appropriate angle with respect to the red light **L1a**, the blue light **L1b**, and the green light **L1c** that are emitted from the light source unit **20B**.

(61) Further, as illustrated in FIG. 8, the second light reception unit **42** includes a first light reception member **421** that receives the red reflection light **L2a** obtained by reflecting the red light **L1a** having the first wavelength and a second light reception member **422** that receives the green reflection light **L2c** obtained by reflecting the green light **L1c** having the third wavelength.

(62) Note that, in the third embodiment, there is adopted a configuration in which one light source unit **20B** is provided. However, a configuration in which a plurality of light source units that respectively emit the three parts of light **L1a**, **L1b**, and **L1c** are provided may be adopted. In other words, a light source unit that emits the blue light **L1b** for detecting the moving direction of the mouse main body **10C** and light source units that emit the red light **L1a** and the green light **L1c** for detecting biometric information about a user may be provided.

(63) According to the third embodiment, the two types of light having different wavelengths (the red light **L1a** and the green light **L1c**) can be reflected by the living body **M** through the second opening **30B**, and the red reflection light **L2a** and the green reflection light **L2c** can separately be received by the first light reception member **421** and the second light reception member **422**, respectively. Thus, the two types of biometric information **D** can be detected efficiently.

Second Modification Example

(64) A second modification example illustrated in FIG. 9 is a configuration obtained by changing the configuration of the second light reception unit **42** in the third embodiment described above. The second light reception unit **42** includes the first light reception member **421** that receives the red light **L1a** having the first wavelength (the red reflection light **L2a**) and the second light reception member **422** that receives the green light **L1c** having the third wavelength (the green reflection light **L2c**).

(65) The first light reception member **421** includes a first color filter **43** that transmits the red light **L1a** having the first wavelength (the red reflection light **L2a**) blocks the blue light **L1b** having the second wavelength (the blue reflection light **L2b**) and the green light **L1c** having the third wavelength (the green reflection light **L2c**), which are illustrated in FIG. 7. The second light reception member **422** includes a second color filter **44** that transmits the green light **L1c** having the third wavelength (the green reflection light **L2c**) and blocks the red light **L1a** having the first wavelength (the red reflection light **L2a**) and the blue light **L1b** having the second wavelength (the blue reflection light **L2b**).

(66) In the second modification example, the first light reception member **421** including the first color filter **43** can only receive the red reflection light **L2a** while blocking the green reflection light **L2c**. The second light reception member **422** including the second color filter **44** can only receive the green reflection light **L2c** while blocking the red reflection light **L2a**.

(67) Some of the embodiments of the present disclosure are described above. However, those embodiments are given merely as examples, and are not intended to limit the scope of the present disclosure. Those embodiments can be carried out in various other forms, and various omission, replacement, and modifications may be made thereto without departing from the gist of the present disclosure. Those embodiments and the modifications thereof are within the scope and the gist of the present disclosure, and are also encompassed within the scope of claims and the scope of the equivalents in the present disclosure.

(68) In the first embodiment described above, there is adopted a configuration in which the moving direction of the mouse main body **10** is detected by using the reflection light reflected by the deflection mirror **51** constituted by a half mirror and the biometric information D about the user is detected by using the light transmitted through the deflection mirror **51**. However, the configuration is not limited thereto. There may be adopted a configuration in which the moving direction of the mouse main body **10** is detected by using the light transmitted through the deflection mirror **51** and the biometric information D about the user is detected by using the light reflected by the deflection mirror **51**.

(69) In the optical mouse **1** (biometric information detection device), the configuration such as a position, orientation, and a shape of each of the components including the light source unit, the detecting unit, the light reception unit, and each of the openings is not limited to the embodiments described above, and may be set as appropriate. That is, it is only required to provide the first opening and the first light reception unit for reflecting the light by the target object P and the second opening and the second light reception unit for reflecting the light by the living body M.

(70) Further, in the present embodiment, the case **30** is provided with the lower case unit **31** including the first surface and the upper case unit **32** including the second surface, but the configuration thereof is not limited thereto.

(71) Further, the control unit **12** may be omitted.

(72) The input device according to one aspect of the present disclosure may include the following configuration.

(73) The input device according to one aspect of the present disclosure is configured to communicate with an information processing device, and includes a light source unit configured to emit light, a case including a first surface including a first opening and configured to emit the light toward a target object through the first opening, and a second surface including a second opening and configured to emit the light toward a living body through the second opening, a first light reception unit configured to receive first light from the target object through the first opening of the first surface, and a second light reception unit configured to receive second light from the living body through the second opening of the second surface.

(74) The input device according to one aspect of the present disclosure may include a deflection unit configured to deflect one part of the light and transmit another part of the light, in which the deflection unit may deflect the one part of the light emitted from the light source unit toward the first opening of the first surface and transmit the another part of the light emitted from the light source unit toward the second opening of the second surface.

(75) In the input device according to one aspect of the present disclosure, the first surface may be a surface of the case that comes into contact with the target object and does not come into contact with the living body, and the second surface may be a surface of the case that comes into contact with the living body and does not come into contact with the target object.

(76) In the input device according to one aspect of the present disclosure, the case may include a lower case unit including the first surface and an upper case unit covering the first surface and including the second surface, and the upper case unit may include the second surface at one end side of the case in a short direction.

(77) In the input device according to one aspect of the present disclosure, the case may include a lower case unit including the first surface and an upper case unit covering the first surface and

including the second surface, and the upper case unit may include a movable portion being movable at one end side of the case in a longitudinal direction and the second surface at the other end side in the longitudinal direction.

(78) The input device according to one aspect of the present disclosure may include an optical member between the light source unit and the deflection unit.

(79) In the input device according to one aspect of the present disclosure, the optical member may be a condensing lens.

(80) In the input device according to one aspect of the present disclosure, the light emitted from the light source unit may include light having a first wavelength and light having a second wavelength different from the first wavelength, and the deflection unit may deflect the light having the second wavelength toward the target object and deflect the light having the first wavelength toward the living body.

(81) In the input device according to one aspect of the present disclosure, the second wavelength may be shorter than the first wavelength.

(82) In the input device according to one aspect of the present disclosure, the light emitted from the light source unit may include light having a third wavelength longer than the first wavelength, the deflection unit may deflect the light having the third wavelength toward the living body, and the second light reception unit may include a first light reception member configured to receive the light having the first wavelength and a second light reception member configured to receive the light having the third wavelength.

(83) In the input device according to one aspect of the present disclosure, the first light reception member may include a first color filter that transmits the light having the first wavelength and blocks the light having the second wavelength and the light having the third wavelength, and the second light reception member may include a second color filter that transmits the light having the third wavelength and blocks the light having the first wavelength and the light having the second wavelength.

(84) The biometric information detection device according to one aspect of the present disclosure may include the following configuration.

(85) The biometric information detection device according to one aspect of the present disclosure includes the input device according to the above-mentioned aspect, the information processing device configured to communicate with the input device, in which the input device is movably provided, and a control unit is provided, the control unit being configured to detect a moving direction of the input device, based on reception of the first light at the first light reception unit, and detect biometric information about a user, based on reception of the second light at the second light reception unit.

(86) In the biometric information detection device according to one aspect of the present disclosure, when the second light reception unit receives the second light, the control unit may detect the moving direction of the input device, based on reception of the first light at the first light reception unit.

(87) In the biometric information detection device according to one aspect of the present disclosure, the control unit may transmit information relating to the biometric information about the user to the information processing device.

(88) The biometric information detection device according to one aspect of the present disclosure may include a detection member configured to detect movement of the input device, and when the detection member detects movement of the input device, the control unit may detect the biometric information about the user, based on reception of the second light at the second light reception unit.

Claims

1. An input device configured to communicate with an information processing device, the input device comprising: a light source unit configured to emit light; a case including: a first surface including a first opening and configured to emit the light toward a target object through the first opening; and a second surface including a second opening and configured to emit the light toward a living body through the second opening; a first light reception unit configured to receive first light from the target object through the first opening of the first surface; a second light reception unit configured to receive second light from the living body through the second opening of the second surface; and a deflection unit configured to deflect one part of the light and transmit another part of the light, wherein the deflection unit deflects the one part of the light emitted from the light source unit toward the first opening of the first surface, and transmits the another part of the light emitted from the light source unit toward the second opening of the second surface, the light emitted from the light source unit includes light having a first wavelength and light having a second wavelength different from the first wavelength, the deflection unit deflects the light having the second wavelength toward the target object and deflects the light having the first wavelength toward the living body, the light emitted from the light source unit includes light having a third wavelength longer than the first wavelength, the deflection unit deflects the light having the third wavelength toward the living body, and the second light reception unit includes a first light reception member configured to receive the light having the first wavelength and a second light reception member configured to receive the light having the third wavelength.
2. The input device according to claim 1, wherein the first surface is a surface of the case configured to come into contact with the target object and not configured to come into contact with the living body and the second surface is a surface of the case configured to come into contact with the living body and not configured to come into contact with the target object.
3. The input device according to claim 1, wherein the case includes: a lower case unit including the first surface; and an upper case unit covering the first surface and including the second surface, and the upper case unit includes the second surface at one end side of the case in a short direction.
4. The input device according to claim 1, wherein the case includes: a lower case unit including the first surface; and an upper case unit covering the first surface and including the second surface, and the upper case unit includes: a movable portion being movable at one end side of the case in a longitudinal direction; and the second surface at the other end side of the case in the longitudinal direction.
5. The input device according to claim 1, further comprising: an optical member between the light source unit and the deflection unit.
6. The input device according to claim 5, wherein the optical member is a condensing lens.
7. The input device according to claim 1, wherein the second wavelength is shorter than the first wavelength.
8. The input device according to claim 1, wherein the first light reception member includes a first color filter configured to transmit the light having the first wavelength and configured to block the light having the second wavelength and the light having the third wavelength, and the second light reception member includes a second color filter configured to transmit the light having the third wavelength and configured to block the light having the first wavelength and the light having the second wavelength.
9. A biometric information detection device, comprising: the input device according to claim 1; and the information processing device configured to communicate with the input device, wherein the input device is movably provided and a control unit is provided, the control unit being configured to detect a moving direction of the input device, based on reception of the first light at the first light reception unit, and detect biometric information about a user, based on reception of the second light at the second light reception unit.
10. The biometric information detection device according to claim 9, wherein when the second

light reception unit receives the second light, the control unit detects the moving direction of the input device, based on the reception of the second light at the second light reception unit.

11. The biometric information detection device according to claim 9, wherein the control unit transmits information relating to the biometric information about the user to the information processing device.

12. The biometric information detection device according to claim 9, further comprising: a detection member configured to detect movement of the input device, wherein when the detection member detects the movement of the input device, the control unit detects the biometric information about the user, based on the reception of the second light at the second light reception unit.
